University of Applied Sciences
HOCHSCHULE EMDEN·LEER

Floor Localization using Nvidia Jetson Nano

using Raspberry Pi lens

Srikanth Nanda

23. Oktober 2025

1. Introduction to Floor Localization
2. Challenges, Solution and Applications
3. Algorithm
4. Hardware Platform: Jetson Nano and Raspberry Pi Camera
5. KDD implementation in Floor Localization
6. Quellen

Introduction to Floor Localization

Introduction I

Image-based floor detection also known as Floor Localization is a key task in mobile robot navigation. It provides the robot with information about the free space in its environment, allowing safe movement and obstacle avoidance. Detecting the floor also supplies structural cues, such as the boundary between floor and walls, which are useful for generating maps of indoor spaces.[LB10]

Introduction II

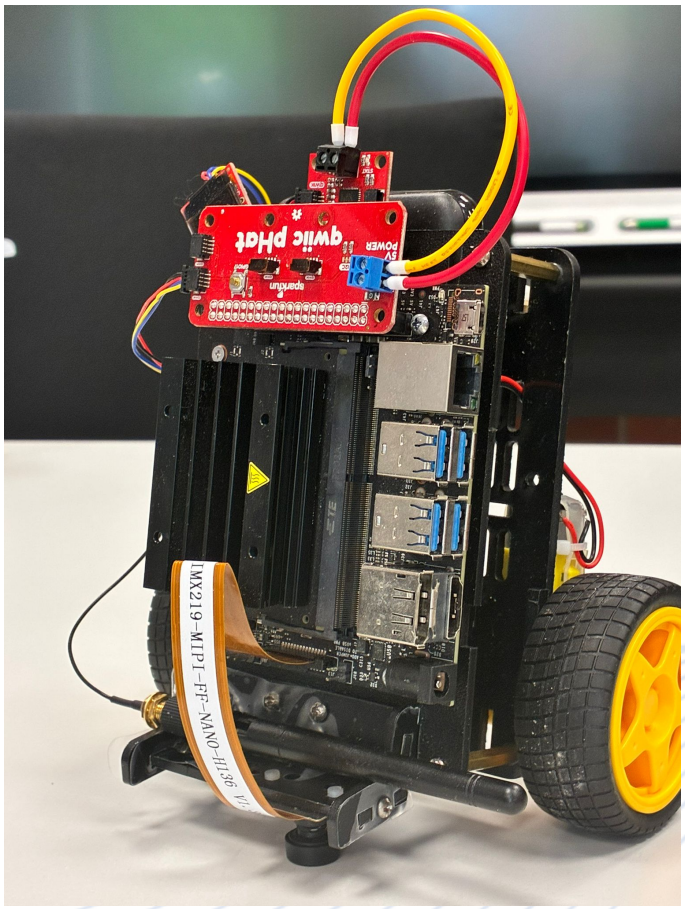


Abbildung: Mobile Robot Implemented using Nvidia Jetson Nano(Top view)

Introduction III

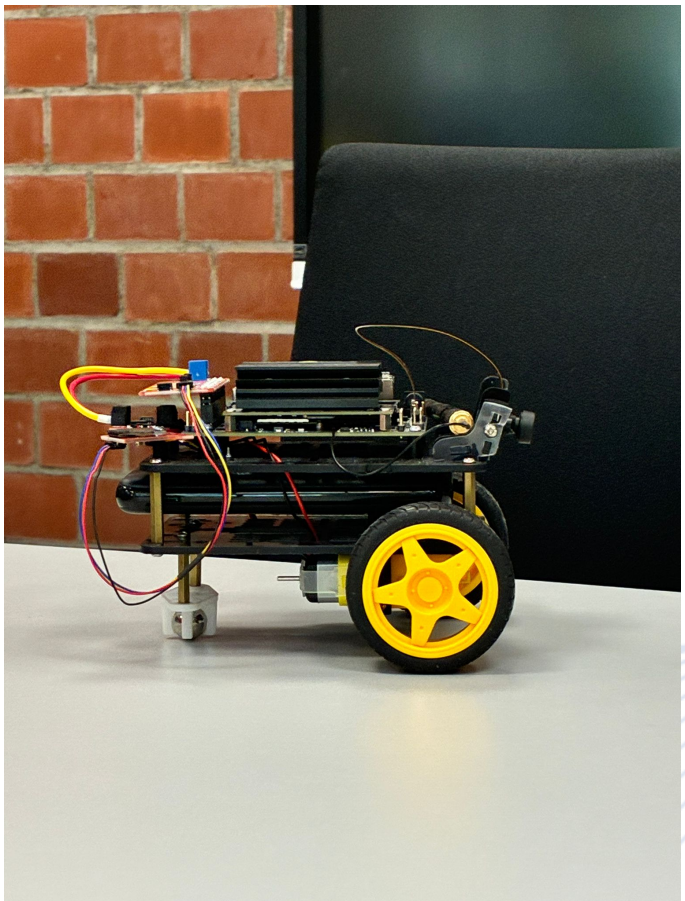


Abbildung: Mobile Robot Implemented using Nvidia Jetson Nano(Side view)

During Localization I

During localization, the identified floor region can be matched with the floor layout in an existing map to support accurate position estimation. Floor detection further enables the calculation of spatial metrics, including the usable area or the dimensions of a room. In this work, these functions are implemented on a four-wheel rover using an NVIDIA Jetson Nano and a Raspberry Pi 2-megapixel camera, building on earlier work in vision-based floor segmentation.[LB10]

Ground Texture Localization I

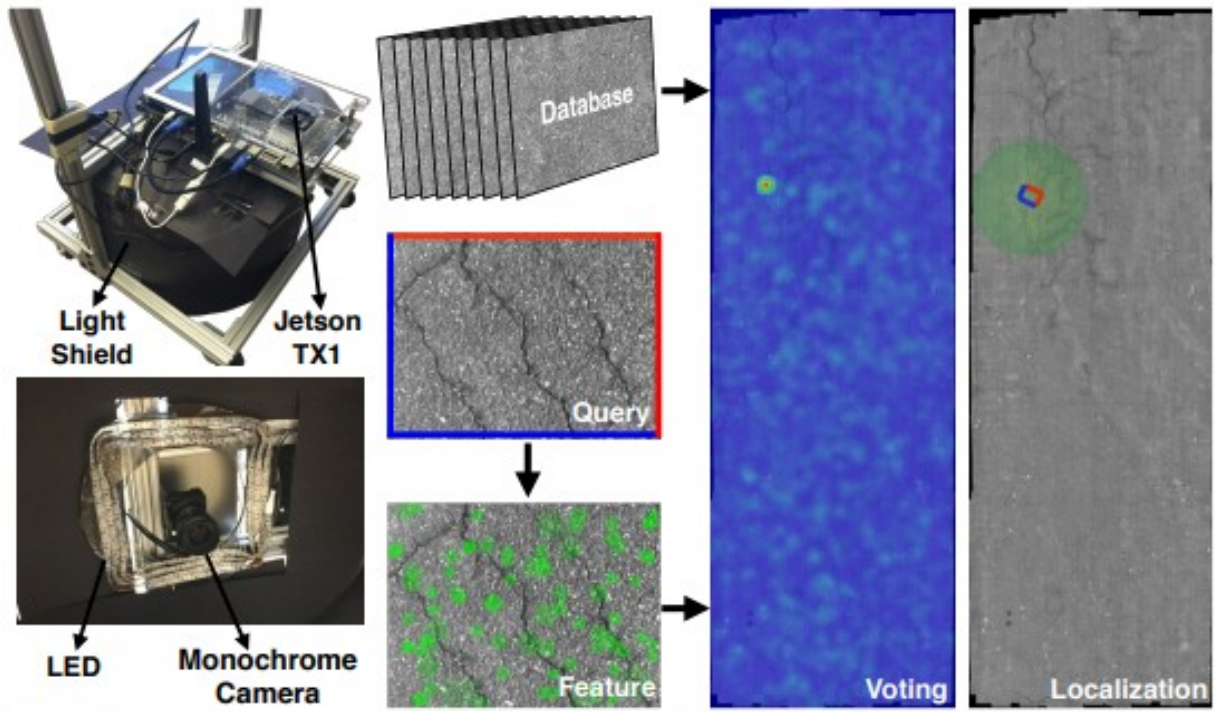


Fig. 1: System overview. Our test robot features an NVIDIA Jetson TX1 development board, which controls a Point Grey mono camera.

Abbildung: Extracting Ground Texture [ZFR19]

Challenges, Solution and Applications

Challenges I

Indoor Navigation Challenges

Indoor positioning systems face several fundamental challenges that make them significantly more complex than their outdoor counterparts:

- **GPS Signal Unavailability**: Global Navigation Satellite Systems (GNSS) signals are severely attenuated or completely blocked in indoor environments, subterranean locations, and dense urban environments [Pas24].
- **Signal Propagation Issues**: Indoor wireless signals suffer from multipath fading, shadowing effects, and reflection interference from walls, furniture, and other obstacles [Aff+23].

Challenges II

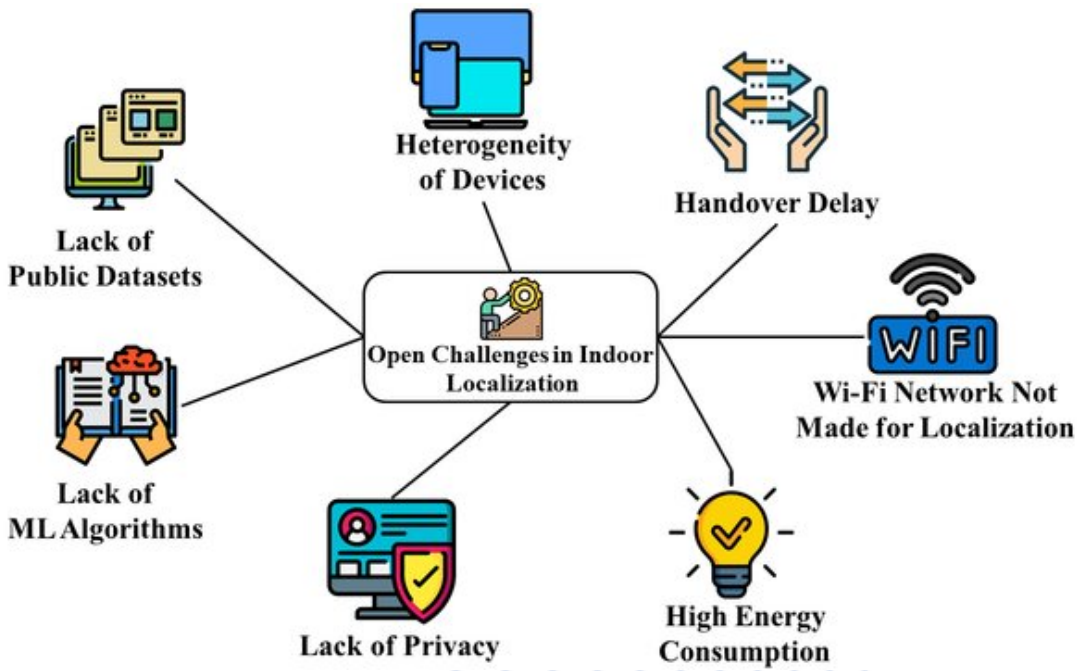


Abbildung: Indoor Navigation Challenges [Cal+21]

Challenges III

- **Dynamic Environment Complexity**: Indoor spaces contain moving objects, varying lighting conditions, and changing layouts that challenge traditional localization algorithms [LR23].
- **Computational Constraints**: Mobile and embedded systems have limited processing power, memory, and energy resources for complex localization algorithms [TKP21].

Solutions I

Computer vision offers robust solutions for indoor localization through several key approaches:

- **Visual Simultaneous Localization and Mapping (VSLAM)**: Constructs environmental maps while tracking camera pose using feature extraction and matching algorithms [BHD24].
- **Deep Learning Integration**: Neural networks enhance feature detection, matching accuracy, and robustness in challenging conditions [BHD25].

Solutions II

- **Image Fingerprinting**: Creates unique visual signatures of locations for position estimation without requiring infrastructure modifications [Wer+11].
- **Multi-Modal Fusion**: Combines visual information with inertial measurement unit (IMU) data for improved accuracy and reliability [Jam+23].

Application I

Real-World Deployment Scenarios
Indoor localization systems using Jetson Nano and computer vision find applications across multiple domains:
Healthcare and Accessibility

- Navigation assistance for visually impaired individuals [Luo+20]
- Hospital patient and equipment tracking [Asi+23]
- Emergency response and evacuation guidance [Bha+24]

Application II

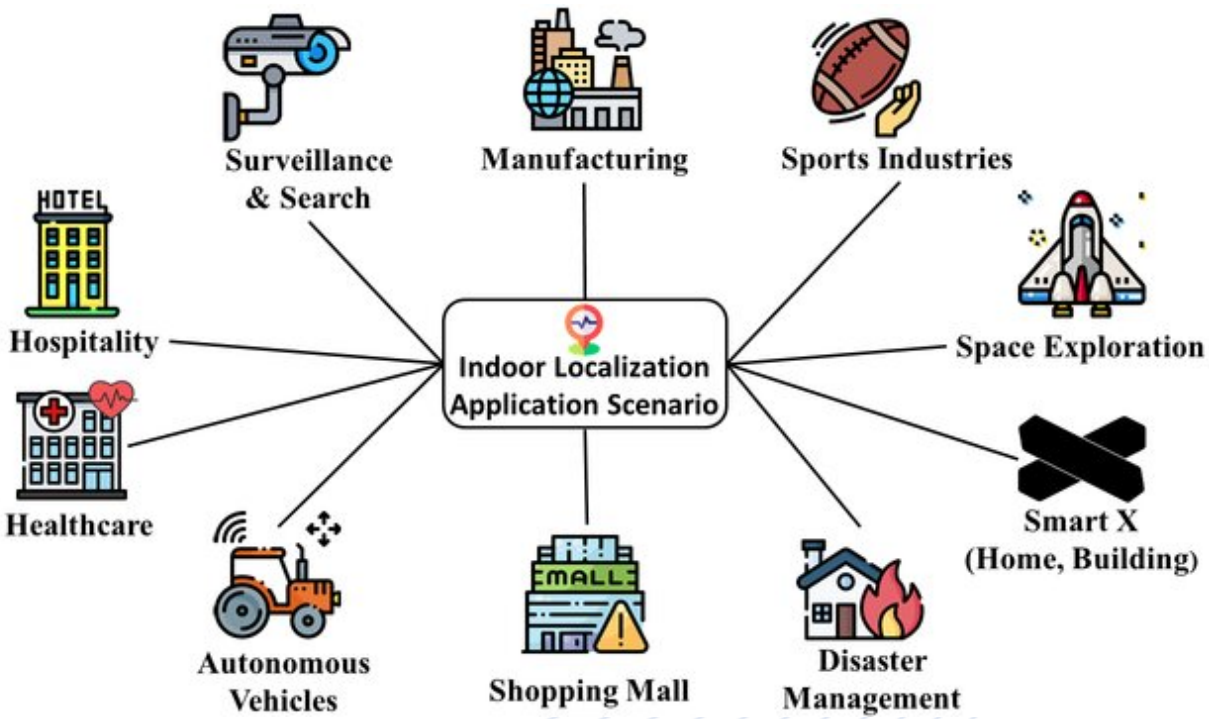


Abbildung: Applications [Cal+21]

Application III

Industrial and Commercial

- Warehouse automation and inventory management. [Sin+23]
- Manufacturing robot navigation and coordination. [CE+20]
- Retail customer analytics and indoor directions. [Sin+23]

Smart Buildings and IoT

- Building management and space utilization.[Wan+24a]
- Asset tracking and security monitoring.[Wan+24a]
- Environmental sensing and control systems.[Luo+20]

Algorithm

Flowchart for Mobile Robot Navigation I

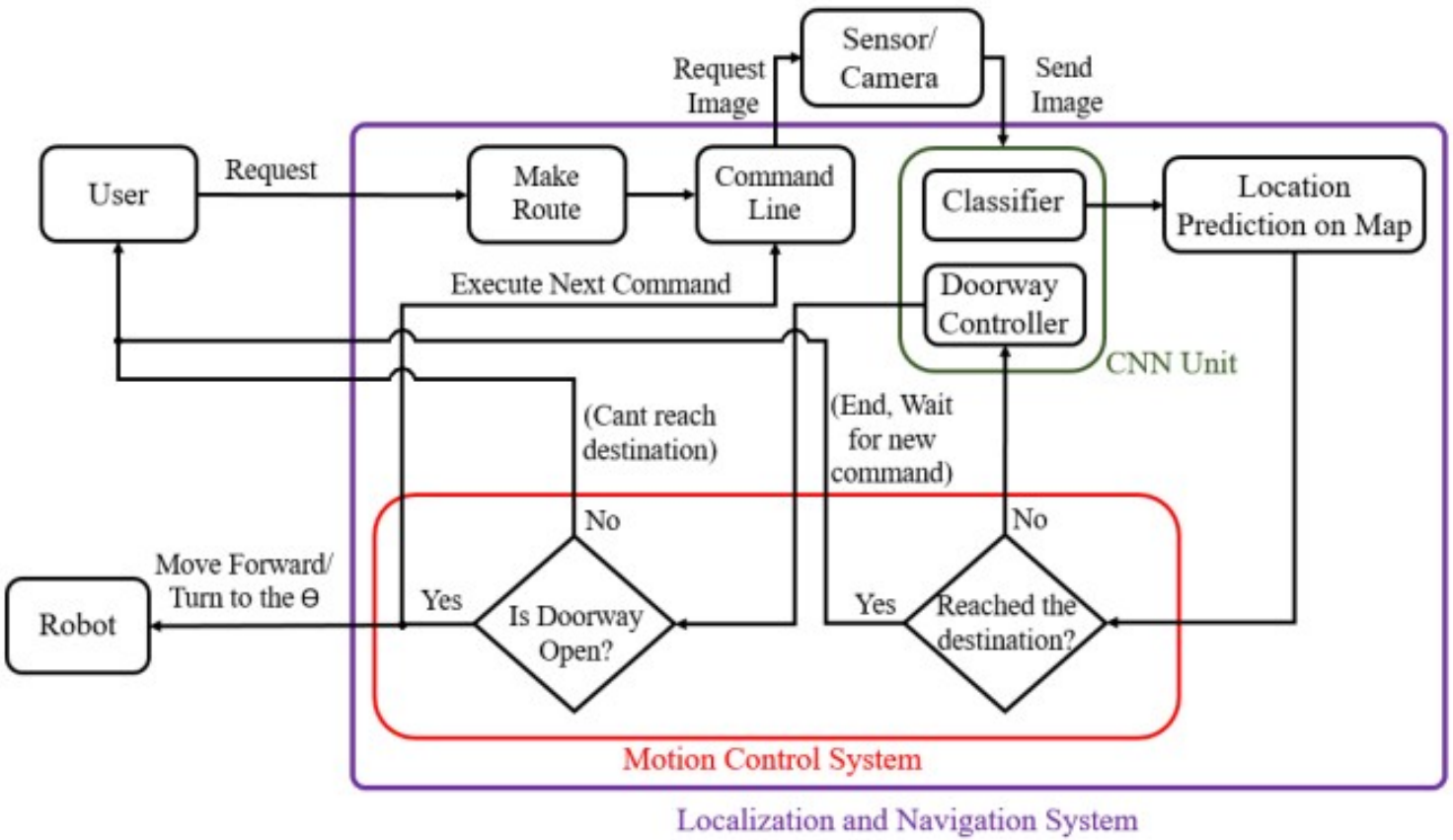


Abbildung: Flowchart for Mobile Robot Navigation[FCW21]

Flowchart for Indoor Localization I

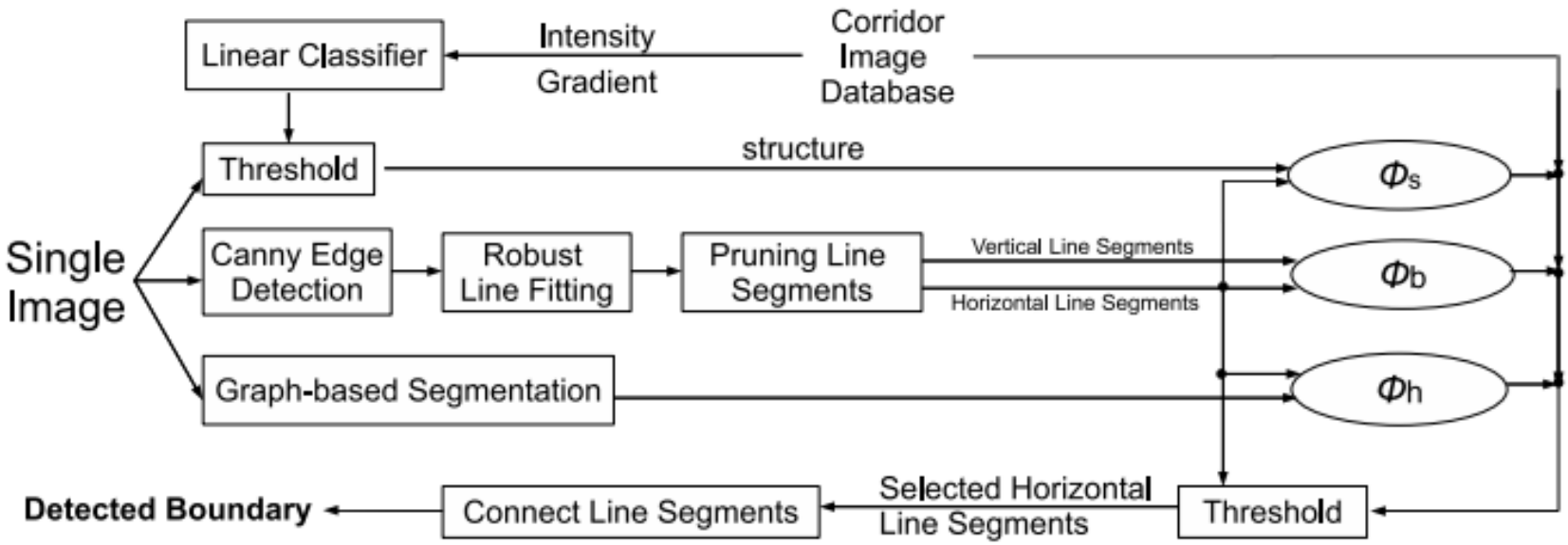


Abbildung: Flowchart for floor detection[LB10]

Mathematical Expression I

To determine the likelihood that a horizontal segment ℓ_h is near this boundary, we compute a weighted sum of scores for three individual visual cues:

$$\Phi_{total}(\ell_h) = w_s\overline{\phi}_s(\ell_h) + w_b\overline{\phi}_b(\ell_h) + w_h\overline{\phi}_h(\ell_h), \tag{1}$$

where w_s, w_b, and w_h are the weights, and $\overline{\phi}_s(\ell_h), \overline{\phi}_b(\ell_h), \overline{\phi}_h(\ell_h)$ are the three individual scores, which are now described.[LB10]

Hardware Platform: Jetson Nano and Raspberry Pi Camera

Nvidia Jetson Nano I

System Architecture

The NVIDIA Jetson Nano developer kit paired with Raspberry Pi Camera Module provides an optimal platform for embedded computer vision applications

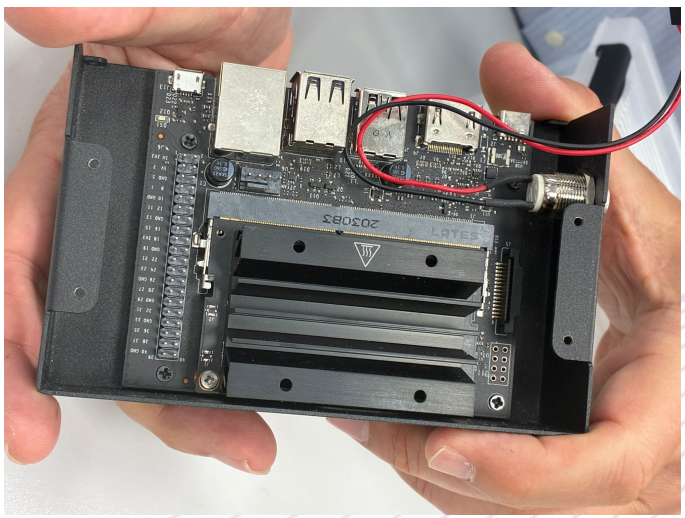


Abbildung: Nvidia Jetson Nano

Nvidia Jetson Nano II

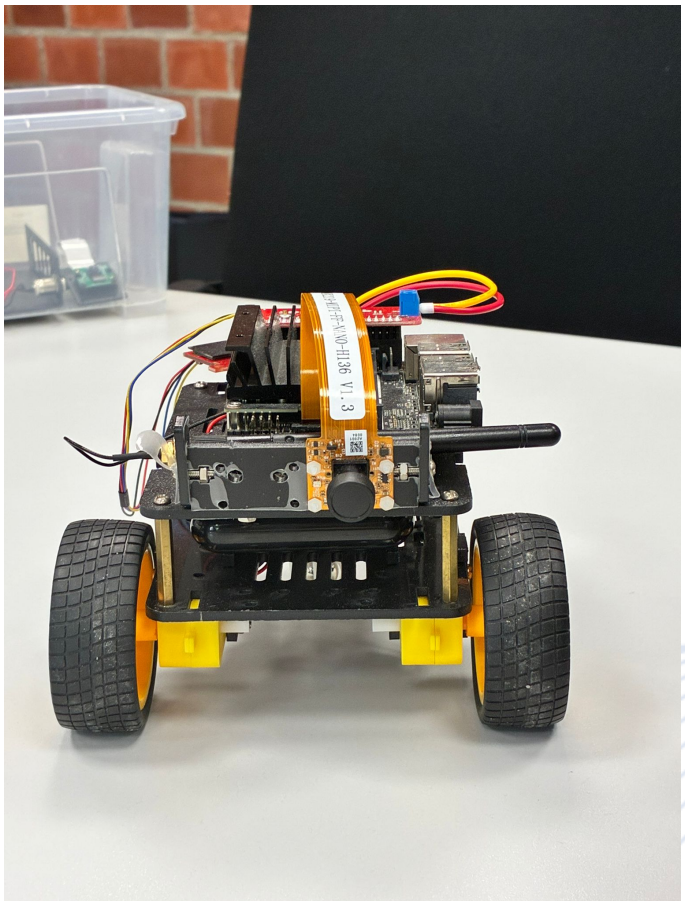


Abbildung: Mobile Robot with Raspberry Pi camera (Front view)

Nvidia Jetson Nano III

Hardware Specifications

- **Jetson Nano**: ARM Cortex-A57 quad-core processor, 128-core Maxwell GPU
- **Camera**: Raspberry Pi Camera Module V2 (8MP, 1080p, Sony IMX219 sensor)
- **Interface**: 15-pin MIPI CSI-2 connection for direct camera integration
- **Processing**: Hardware-accelerated computer vision with CUDA support

KDD implementation in Floor Localization

Machine Learning Pipeline I

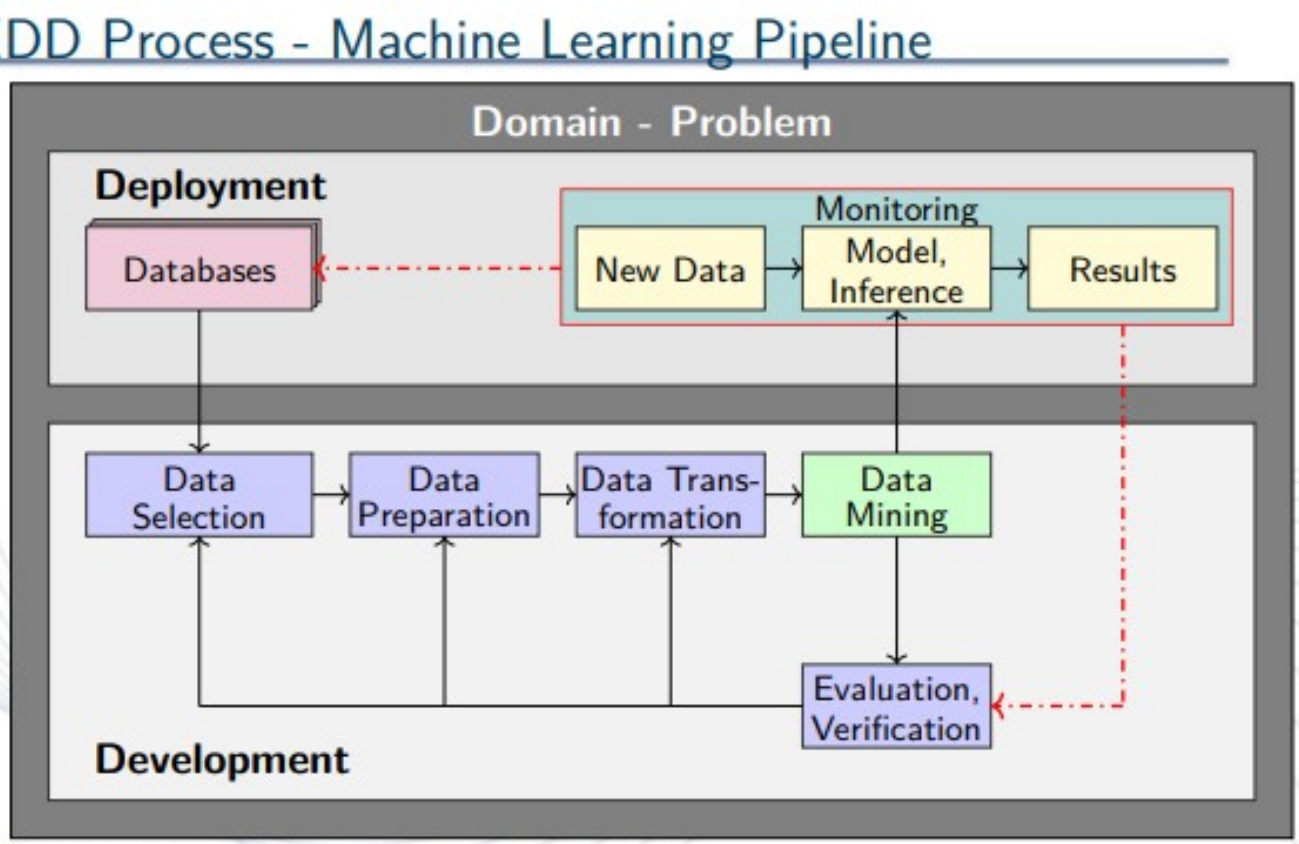


Abbildung: KDD Process - ML Pipeline

KDD Implementation for Floor Localization I

- Problem understanding: robust indoor floor localization for a mobile robot to support safe navigation and obstacle avoidance; real-time inference on embedded hardware (Jetson Nano) from a Raspberry Pi Camera stream under edge-compute constraints [Luo+20; CE+20].
- Data understanding and database: sources are RGB frames (CSI-2), optional IMU (accelerometer/gyroscope), environment metadata (lighting, floor material), and ground truth via annotated floor boundaries and pose labels (x, y, θ); streams feature high velocity, varied textures, and illumination changes [FCW21; ZFR19].

KDD Implementation for Floor Localization II

- Data selection: retain frames with adequate texture/edges, acceptable shutter time, low blur; select edge maps, line segments, vanishing-point cues, optical flow, and IMU channels for fusion [Wan+24b; Yu+18].
- Data preparation: calibrate/undistort camera, normalize exposure and white balance, deblur/denoise; time-synchronize camera and IMU, remove outliers, interpolate short dropouts; split by location/session to prevent leakage [Lu+24; Luo+24].

KDD Implementation for Floor Localization III

- Data transformation: engineer features (Canny, line detection, graph/superpixel cues, optical-flow summaries), normalize image/IMU signals and pose units, augment with brightness/contrast jitter, rotations, crops, perspective warps [Xu+24b; Xie+24; Xu+24a].
- Data mining (modeling): use a lightweight CNN for floor/non-floor and boundary likelihoods optimized with TensorRT on Jetson Nano; fuse CNN measurements with IMU via EKF/VIO for drift reduction; output (x, y, θ) and uncertainty, optionally updating a floor/occupancy map [Liu+24; Kim+23; Zha+24].

KDD Implementation for Floor Localization IV

- Evaluation and verification: report ATE/RPE, boundary IoU/F1, route success rate, and on-device latency/FPS; validate with k-fold across rooms/floors, stress tests (illumination/occlusions), and ablations (vision-only vs. vision+IMU) [Lee+20; He+24].
- Deployment: export to ONNX and TensorRT with fixed input (e.g., 256×256), integrate pose/uncertainty into the motion controller to issue velocity/turn commands with safety stops [Hu+22; Qin+23].
- Monitoring and maintenance: track pose error vs. landmarks, FPS, dropped frames, IMU bias; detect data drift, log failures for replay, periodically fine-tune and re-check calibration.

Vielen Dank

für Ihre Aufmerksamkeit

Quellen

Quellen I

Nachfolgend werden die Quellen der Bilder angegeben, die für diese Präsentation in ihrer ursprünglichen Form oder modifiziert verwendet worden sind.

[Aff+23] A. Affan u. a. "Machine-Learning-Based Indoor Localization under Challenging Conditions Using VLC Systems". In: *PMC* 10 (2023), S. 255998. DOI: 10.3390/s23115319.

[Asi+23] Muhammad Asif u. a. "Face mask detection and counting using you only look once algorithm with Jetson Nano and NVIDIA giga texel shader extreme". In: *International Journal of Advanced Intelligence* (2023). URL: https://ijai.iaescore.com/index.php/IJAI/article/view/21815.

Quellen II

[Bas+23] D. C. S. Basavaraddi u. a. "A Project Work on Computer Vision Based Indoor Navigation System". In: *International Journal of Advanced Research in Computer Science and Engineering Technology* 10.5 (2023), S. 561. DOI: 10.17148/IARJSET.2023.10561.

[Bha+24] Simmu Bhangu u. a. "Vision-based indoor localization of nano drones in controlled environment with its applications". In: (2024). URL: https://arxiv.org/abs/2412.08757.

[BHD24] M. Bamdad, H.-P. Hutter und A. Darvishy. "InCrowd-VI: A Realistic Visual-Inertial Dataset for Evaluating SLAM in Indoor Pedestrian-Rich Spaces for Human Navigation". In: *arXiv preprint arXiv:2411.14358* (2024). DOI: 10.48550/arXiv.2411.14358.

Quellen III

[BHD25] M. Bamdad, H.-P. Hutter und A. Darvishy. "Deep Learning-Powered Visual SLAM Aimed at Assisting Visually Impaired Navigation". In: *SCITEPRESS Conference Proceedings*. 2025. DOI: 10.5220/0013338200003822.

[CaI+21] Luigi Caltagirone u. a. "LIFELONG Learning for Advanced Visual Perception of Environmental Scenes". In: *IEEE Access* 9 (2021), S. 118980–118993. DOI: 10.1109/ACCESS.2021.3111083. URL: https://doi.org/10.1109/ACCESS.2021.3111083.

[CE+20] Eddy Camacho-Escobar u. a. "Embedded Mobile ROS Platform for SLAM Application with RGB-D Cameras". In: *IEEE Access* (2020). DOI: 10.1109/ACCESS.2020.2994245. URL: https://ieeexplore.ieee.org/document/9208310/.

Quellen IV

[Coo+16] A. Cooper u. a. "An Indoor Positioning System Facilitated by Computer Vision". In: *IEEE Conference Proceedings*. 2016. DOI: 10.1109/MCMSNJ.2016.7890234.

[FCW21] Farzin Foroughi, Zonghai Chen und Jikai Wang. "A CNN-Based System for Mobile Robot Navigation in Indoor Environments via Visual Localization with a Small Dataset". In: *World Electric Vehicle Journal* 12.3 (2021), S. 134. DOI: 10.3390/wevj12030134. URL: https://doi.org/10.3390/wevj12030134.

[He+24] Dehong He u. a. "Point-line Visual-Inertial Odometry Systems under Short and Long Line Segment Fusion". In: *Transactions of the Institute of Measurement and Control* (2024). DOI: 10.1177/01423312241285949.

Quellen V

[Hu+22] Jianming Hu u. a. "EMA-VIO: Deep Visual-Inertial Odometry with External Memory Attention". In: (2022). URL: https://arxiv.org/pdf/2209.08490.pdf.

[Jam+23] A. James u. a. "Sensor Fusion for Autonomous Indoor UAV Navigation in Confined Spaces". In: *IEEE International Conference on Smart Technologies*. 2023. DOI: 10.1109/ICST59744.2023.10460820.

[Kim+23] Jae-Hean Kim u. a. "PLI-VIO: Real-time Monocular Visual-inertial Odometry Using Point and Line Interrelated Features". In: *International Journal of Control, Automation and Systems* (2023). DOI: 10.1007/s12555-022-0210-9.

Quellen VI

[LB10] Yinxiao Li und Stanley T. Birchfield. "Image-Based Segmentation of Indoor Corridor Floors for a Mobile Robot". In: *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2010, S. 837–843. DOI: 10.1109/IROS.2010.5652818.

[Lee+20] Kyuman Lee u. a. "Latency Compensated Visual-Inertial Odometry for Agile Autonomous Flight". In: *Sensors* 20.8 (2020), S. 2209. DOI: 10.3390/s20082209.

[Liu+22] L. Liu u. a. "Indoor Fingerprinting Positioning System Using Deep Learning". In: *Sensors and Materials* 34.8 (2022), S. 3059. DOI: 10.18494/SAM3922.

[Liu+24] Yuxuan Liu u. a. "Load Balanced Sparse Bundle Adjustment Accelerator for CNN-Based Visual-Inertial Odometry on FPGA". In: *IEEE Transactions on Circuits and Systems I* (2024). DOI: 10.1109/TCSI.2024.3506861.

Quellen VII

[LR23] N. Linus und E. Rueckert. "Understanding Why SLAM Algorithms Fail in Modern Indoor Environments". In: *arXiv preprint arXiv:2305.05313* (2023). DOI: 10.48550/arXiv.2305.05313.

[Lu+24] Dongyue Lu u. a. "RMSC-VIO: Robust Multi-Stereoscopic Visual-Inertial Odometry for Local Visually Challenging Scenarios". In: *IEEE Robotics and Automation Letters* (2024). DOI: 10.1109/LRA.2024.3396100.

[Luo+20] Yin Luo u. a. "A Comprehensive Survey of Indoor Localization Methods Based on Computer Vision". In: *Sensors* (2020). DOI: 10.3390/s20092641. URL: https://www.mdpi.com/1424-8220/20/9/2641.

Quellen VIII

[Luo+24] Yonghan Luo u. a. "VIO-DualProNet: Visual-Inertial Odometry with Learning Based Process Noise Covariance". In: (2024). URL: http://arxiv.org/pdf/2308.11228.pdf.

[NG23] M. Nabati und S. A. Ghorashi. "A Real-Time Fingerprint-Based Indoor Positioning Using Deep Learning and Preceding States". In: *Expert Systems with Applications* 213 (2023), S. 119078. DOI: 10.1016/j.eswa.2022.119078.

[Pas24] S. Pasricha. "AI and Machine Learning Driven Indoor Localization and Navigation with Mobile Embedded Systems". In: *arXiv preprint arXiv:2408.04797* (2024). DOI: 10.48550/arXiv.2408.04797.

Quellen IX

[Qin+23] Tong Qin u. a. “PIEKF-VIWO: Visual-Inertial-Wheel Odometry using Partial Invariant Extended Kalman Filter”. In: (2023). URL: https://arxiv.org/pdf/2303.07668.pdf.

[Sin+23] Abhishek Singh u. a. “Checkout System with Object Detection using NVIDIA Jetson Nano and Raspberry Pi”. In: *IEEE Conference on Computer Vision*. 2023. URL: https://ieeexplore.ieee.org/document/10383181/.

[TKP21] S. Tiku, P. Kale und S. Pasricha. “QuickLoc: Adaptive Deep-Learning for Fast Indoor Localization with Mobile Devices”. In: *arXiv preprint arXiv:2104.07521* (2021). DOI: 10.48550/arXiv.2104.07521.

Quellen X

[Wan+24a] Jian Wang u. a. "Vision-Based UAV Detection and Localization to Indoor Positioning System". In: *Sensors* (2024). DOI: 10.3390/s240413121. URL: https://www.mdpi.com/1424-8220/24/13/4121/pdf.

[Wan+24b] Xuewei Wang u. a. "SP-VIO: Robust and Efficient Filter-Based Visual Inertial Odometry with State Transformation Model and Pose-Only Visual Description". In: (2024). URL: https://arxiv.org/abs/2411.07551.

[Wer+11] M. Werner u. a. *Indoor Positioning Using Smartphone Camera*. Techn. Ber. Technical University of Munich, 2011. DOI: 10.14459/2011md1234567.

[Xie+24] Weijian Xie u. a. "360-VIO: A Robust Visual–Inertial Odometry Using a 360° Camera". In: *IEEE Transactions on Industrial Electronics* (2024). DOI: 10.1109/TIE.2024.3351596.

Quellen XI

[Xu+24a] Jianbo Xu u. a. "RD-VIO: Robust Visual-Inertial Odometry for Mobile Augmented Reality in Dynamic Environments". In: *IEEE Transactions on Visualization and Computer Graphics* (2024). DOI: 10.1109/TVCG.2024.3372093.

[Xu+24b] Jinhong Xu u. a. "SG-VIO: Monocular Visual-Inertial Odometry With Tightly Coupled Structural Lines and Gravity to Avoid Degeneracy". In: *IEEE Internet of Things Journal* (2024). DOI: 10.1109/JIOT.2024.3481356.

[Yu+18] Hongshan Yu u. a. "Trifo-VIO: Robust and Efficient Stereo Visual Inertial Odometry using Points and Lines". In: (2018). URL: https://arxiv.org/pdf/1803.02403v1.pdf.

Quellen XII

[ZFR19] Linguang Zhang, Adam Finkelstein und Szymon Rusinkiewicz. "High-Precision Localization Using Ground Texture". In: *arXiv preprint arXiv:1710.10687* (2019). DOI: 10.48550/arXiv.1710.10687. URL: https://arxiv.org/abs/1710.10687.

[Zha+24] Yidi Zhang u. a. "PGD-VIO: An Accurate Plane-Aided Visual-Inertial Odometry with Graph-Based Drift Suppression". In: (2024). URL: https://arxiv.org/html/2407.17709v1.

[ZLC21] Z. Zhang, M. Lee und S. Choi. "Deep-Learning-Based Wi-Fi Indoor Positioning System Using Continuous CSI of Trajectories". In: *Sensors* 21.17 (2021), S. 5850. DOI: 10.3390/s21175850.